Industrial Internship Report

SATHYABAMA

INSTITUTE OF SCIENCE AND TECHNOLOGY

(DEEMED TO BE UNIVERSITY)

Project Title

Crop Yield Prediction Using Machine Learning

Prepared By

SUBHANI SHAIK

Executive Summary

This internship developed a crop yield prediction system using Random Forest Regression.

Preface

The internship improved practical knowledge of Python, Pandas, NumPy, Machine Learning and data analysis.

Introduction

The project predicts crop yield using crop, state and cultivation cost features.

Problem Statement

Estimate crop yield accurately using historical agricultural data.

Existing and Proposed Solution

Existing methods are manual. The proposed ML model automates prediction with higher accuracy.

Proposed Design

Data Collection -> Preprocessing -> Label Encoding -> Train/Test Split -> Random Forest -> Prediction.

Performance Test

R2 Score: 0.9463
MAE: 28.44
MSE: 4852.74

My Learnings

Learned preprocessing, visualization, model training, evaluation and model serialization.

Future Scope

Include weather, rainfall and soil data and deploy as a web application.